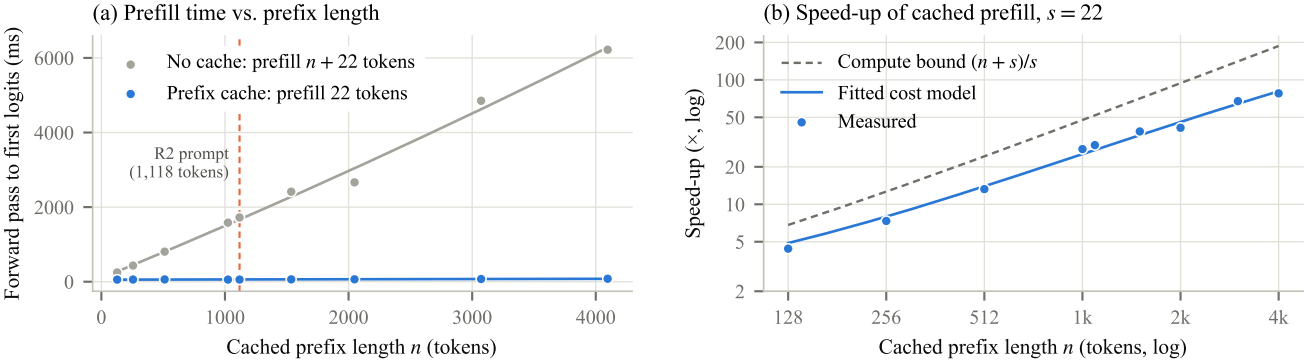
(a) Prefill time vs. prefix length
Forward pass to first logits (ms)
6000
4000
2000
0
No cache: prefill $n + 22$ tokens
Prefix cache: prefill 22 tokens
R2 prompt
(1,118 tokens)
0
1000
2000
3000
4000
Cached prefix length n (tokens)
(b) Speed-up of cached prefill, $s = 22$
Speed-up (×, log)
200
100
50
20
10
5
2
Compute bound $(n + s)/s$
Fitted cost model
Measured
128
256
512
1k
2k
4k
Cached prefix length n (tokens, log)